Models and measurements

An observation model describes how a scientific process is measured.

Underlying process
$x(t)$

→

Observation model
$p(y \mid x, \theta_{\text{obs}})$

→

Observed data
y

Established: health

Disease states
Detection / reporting delay
Cases / admissions / serology

Established: ecology

Wildlife infection / plankton dynamics
Sampling / spatial aliasing
Assays / counts / surveys

Established: incomplete data

Traits / latent structure
Missingness / coding / sampling
Assemblages / language / genetics

Developing: monitoring design

Sampling schedule / modality
Instrument settings / reference data
Cost / decision utility

Established denotes existing published or operational evidence; developing denotes an active methods direction. Every application still needs domain-specific assumptions and independent validation.